TASK 3

Interpretability of Linear Models

Reliable Models

EDM - GCD

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1. Introduction

1.1. Context and Motivation

The interpretability of machine learning models is today one of the cornerstones of responsible AI. In fields such as medicine, finance, or public administration, predictive accuracy alone is not enough: it is essential to understand **why** a model makes each decision. This understanding allows practitioners to detect biases, build trust with end users, and comply with increasingly demanding regulatory frameworks.

In this exercise we apply multiple linear regression to predict daily bicycle rentals. Linear regression offers a key advantage over more complex models: each coefficient has a direct and quantifiable meaning, making it a fundamental reference point in the field of Explainable AI (XAI).

1.2. Objectives

- Understand the principles of interpretability in linear regression models.
- Learn to interpret model coefficients rigorously and communicate them clearly.
- Evaluate the validity of interpretations through statistical diagnostics.
- Visualise results effectively using weight plots and effect plots.
- Identify the inherent limitations of linear models in real-world settings.

1.3. The Problem: Predicting Bicycle Rentals

We use the *Bike-Sharing* dataset from the Capital Bikeshare system in Washington D.C. (2011-2012). This dataset contains daily aggregated information about bicycle rentals, along with weather conditions and calendar features associated with each day.

Research question: What are the factors that influence the number of bicycles rented each day, and what is the quantitative impact of each of those factors?

Target variable

- **cnt**: Total number of bicycles rented on a given day (casual + registered users).

Independent variables

- **season**: Season of the year (1: spring, 2: summer, 3: fall, 4: winter).
- **holiday**: Public holiday indicator (0/1).
- **workingday**: Working day indicator (0/1).
- **weathersit**: Weather condition (1: clear, 2: misty/cloudy, 3: light rain/snow, 4: heavy rain/storm).
- **temp**: Normalised temperature (0-1).
- **atemp**: Normalised apparent temperature (0-1).
- **hum**: Normalised humidity (0-1).

- **windspeed**: Normalised wind speed (0-1).

1.4. Initial Hypotheses

Before fitting the model, we formulated the following hypotheses based on common sense and domain knowledge:

- **Positive effect of temperature:** Warmer days should encourage bicycle use.
- **Negative effect of rain:** Rain and storms should significantly reduce rentals.
- **Effect of day type:** Differences are expected between working days, weekends, and holidays.
- **Seasonal effect:** Usage should be higher in summer and lower in winter.
- **Inter-annual trend:** Rentals likely grew from 2011 to 2012 as the service gained popularity.

2. Data Preprocessing

2.1. Loading and Initial Exploration

The original dataset contains **731 observations** and 16 variables. After loading, we performed a general inspection. The target variable *cnt* has a mean of 4,504 daily rentals with a standard deviation of 1,937, ranging from a minimum of 22 to a maximum of 8,714. This wide spread already hints at the strong influence of seasonal and meteorological factors.

2.2. Feature Engineering

The exercise requires working with 11 specifically constructed features derived from the original dataset. Each transformation is described below.

One-hot encoding of season

The variable *season* is categorical with four levels. To include it in the linear model, one-hot encoding is applied, generating three binary variables (winter acts as the implicit reference category):

- *season1* = 1 if spring, 0 otherwise.
- *season2* = 1 if summer, 0 otherwise.
- *season3* = 1 if fall, 0 otherwise.

Binary weather features

Two binary indicators are created from *weathersit* to simplify the interpretation of atmospheric conditions:

- **MISTY**: Takes value 1 when conditions are misty or cloudy (weathersit = 2).
- **RAIN**: Takes value 1 when there is light rain, snow, or a storm (weathersit = 3 or 4).

Denormalisation of continuous variables

The variables *temp*, *hum*, and *windspeed* were originally normalised to the [0, 1] range. They are denormalised to their real-world scales to drastically improve interpretability: a coefficient expressed in degrees Celsius or km/h is intuitive, whereas one on a [0, 1] scale is opaque.

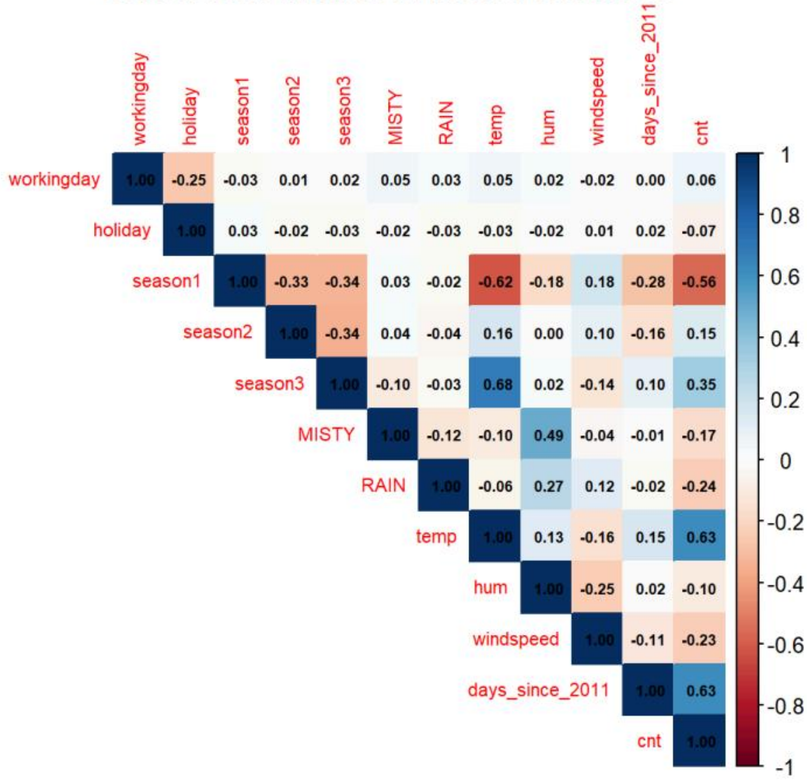
- *temp*: multiplied by 41 (degrees Celsius).
- *hum*: multiplied by 100 (percentage).
- *windspeed*: multiplied by 67 (km/h).

Long-term trend: days_since_2011

A variable is constructed that counts the number of days elapsed since 1 January 2011. This variable acts as a **proxy for the long-term trend**, capturing the organic growth of the service independently of seasonal factors.

2.3. Correlation Analysis

Correlation matrix of model variables



Before fitting the model, a pairwise correlation matrix was computed among all 11 features and the target variable. The most relevant findings are:

- **days_since_2011** and **temp** show the strongest positive correlations with cnt (0.63 each), anticipating that they will be the most influential predictors.
- **season1** and **RAIN** exhibit negative correlations, consistent with the intuition that early spring and rainfall reduce rentals.
- The correlation between temp and season1 is notable (-0.62), signalling potential partial collinearity that may affect the interpretation of individual coefficients.

3. The Linear Model

3.1. Model Fitting

A **multiple linear regression** model is fitted that estimates the daily number of bicycle rentals (*cnt*) as a linear combination of the 11 engineered features. The model specification in R is:

```
model <- lm(cnt ~ workingday + holiday + season1 + season2 + season3 +
                  MISTY + RAIN + temp + hum + windspeed + days_since_2011,
            data = data)
```

Goodness of fit: The model achieves $R^2 = 0.7936$, meaning it explains approximately **79% of the variance** in daily rentals. The F-statistic is highly significant ($p < 2.2e-16$), confirming that the set of predictors jointly explains a meaningful portion of the outcome. The residual standard error is 886.9 rentals.

3.2. Model Coefficients

The following table presents the estimated coefficients, their standard errors, t-statistics, and associated p-values:

Variable	Coefficient	Std. Error	t-statistic	p-value
(Intercept)	1939.3684	275.9336	7.028	< 0.001 ***
workingday	124.9209	73.2666	1.705	0.089 .
holiday	-686.1154	203.3015	-3.375	< 0.001 ***
season1 (Spring)	-425.6029	110.8199	-3.840	< 0.001 ***
season2 (Summer)	473.7153	109.9467	4.309	< 0.001 ***
season3 (Fall)	-287.3874	134.2245	-2.141	0.033 *
MISTY	-379.3985	87.5532	-4.333	< 0.001 ***
RAIN	-1901.5399	223.6400	-8.503	< 0.001 ***
temp	126.9110	8.0740	15.718	< 0.001 ***
hum	-17.3772	3.1694	-5.483	< 0.001 ***
windspeed	-42.5135	6.8917	-6.169	< 0.001 ***
days_since_2011	4.9264	0.1728	28.507	< 0.001 ***

3.3. Interpretation of Key Coefficients

Below we provide a plain-language interpretation of the four requested coefficients, holding all other variables constant in each case.

Temperature (temp)

An increase of **one degree Celsius** in temperature increases the expected number of daily rentals by **126.91 bicycles**. This is one of the most robust effects in the model ($t = 15.72$, $p < 0.001$) and reflects that users are highly sensitive to thermal comfort. Hot summer days can contribute more than 2,000 additional rentals compared to a cold winter day.

Wind speed (windspeed)

Each increase of **1 km/h** in wind speed reduces the expected number of daily rentals by **42.51 bicycles**. The effect is statistically significant ($t = -6.17$, $p < 0.001$) but moderate in magnitude: even on the windiest days in the dataset, this factor dampens rentals far less than a rainy day or the positive boost from high temperatures.

Working day (workingday)

Transitioning from a non-working day to a **working day** is associated with an increase of **124.92 rentals**. However, this coefficient only marginally reaches conventional statistical significance ($p = 0.089$), suggesting that the effect of day type is weaker or less consistent than expected — possibly because the pattern of use varies by season or month.

Holiday (holiday)

A public holiday reduces expected rentals by **686.12 bicycles** compared to an ordinary day ($p < 0.001$). This drop, notably larger than the working-day increment, reveals that the user base is dominated by **regular commuters** who do not use the service on holidays, and that recreational cyclists do not compensate for this loss.

3.4. Simultaneous Change of Two Features

Linear models assume **additivity**: the combined effect of changing two features at once is simply the sum of their individual effects. For example, if a day is a working day (+125 rentals) and wind speed increases by 5 km/h (-213 rentals), the total predicted change is +125 - 213 = -88 rentals.

This property greatly simplifies interpretation, but it also represents a **fundamental limitation**: the model cannot capture **interaction effects** — situations where the impact of one variable depends on the value of another. For instance, strong winds may deter recreational cyclists on weekends more than commuters on working days; a purely linear model is blind to such conditional relationships.

3.5. Limitations of the Interpretation

Although linear model coefficients are intuitive, their interpretation can be misleading in several scenarios:

- **Confounding:** A coefficient captures the marginal association between a feature and the outcome, but this association may be driven by an unmeasured third variable. For example, days_since_2011 absorbs both the growth trend of the service and residual seasonal patterns.
- **Collinearity:** Correlated features share explanatory power, making individual coefficients unstable and harder to interpret in isolation. The correlation between temp and season1 (-0.62) is a clear example in this dataset.
- **Model mis-specification:** If the true relationship is non-linear (e.g., the effect of temperature plateaus at very high values), the linear model will produce biased and misleading coefficients.
- **Outliers and influential observations:** A single extreme observation can substantially shift coefficient estimates.

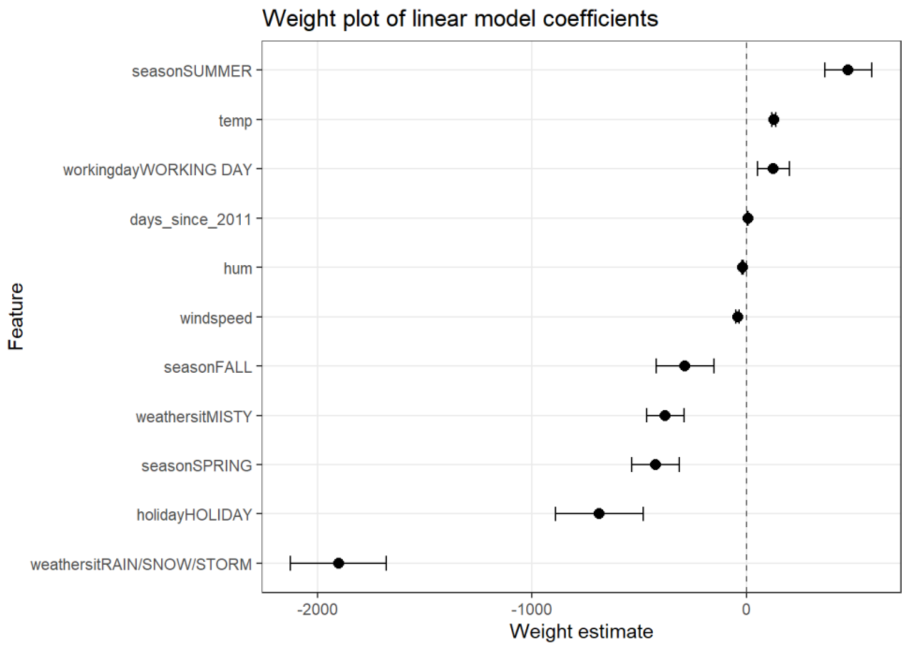
4. Visual Interpretation

Linear regression is intrinsically interpretable: it learns a fixed linear relationship between each feature and the target. However, reading a table of numbers is rarely effective when communicating results to a non-technical audience. The visual tools presented in this section complement the numerical analysis and facilitate the communication of findings.

4.1. Weight Plot

The weight plot displays each feature's estimated coefficient as a point, with a horizontal bar representing its standard error. Features whose confidence interval does not cross zero are statistically significant. The position of the point indicates the direction and magnitude of the effect.

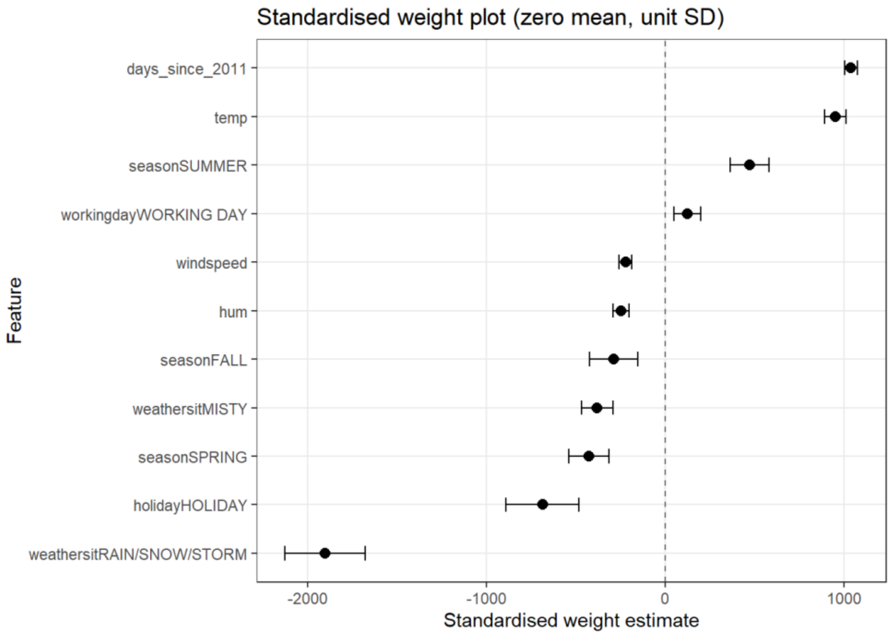
Unstandardised weight plot



When coefficients are shown on their original scales, **weathersitRAIN/SNOW/STORM** stands out visually with the most pronounced negative effect (around -1,901), followed by **holidayHOLIDAY**. On the positive side, **seasonSUMMER** leads with approximately +474, followed by *temp* and *workingday*.

However, this visual comparison is **potentially misleading**: the RAIN coefficient captures a discrete shift between good weather and a storm, whereas the temp coefficient represents the effect of a single degree Celsius. A large bar does not necessarily imply a stronger real-world influence on the business.

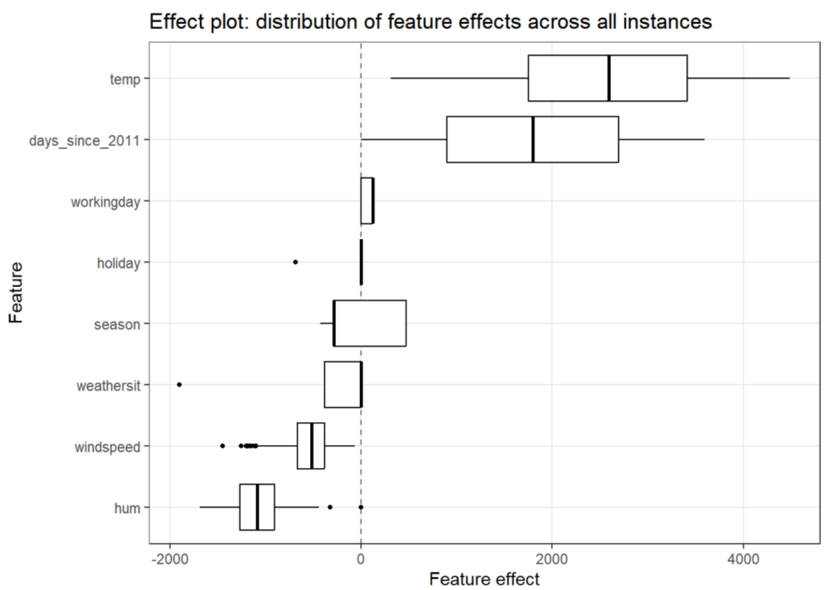
Standardised weight plot



To address the scale problem, all continuous features are standardised to zero mean and unit standard deviation before refitting the model. After standardisation, each coefficient represents the effect of a **one-standard-deviation change** in the predictor, placing comparisons across features on equal footing.

With this representation, **days_since_2011** and **temp** emerge as the two strongest positive drivers of daily rentals, while **RAIN** consolidates as the strongest negative driver. This ranking is far more reliable for assessing the relative importance of features than the raw coefficient plot.

4.2. Effect Plot



The weight plot shows the model's learned parameters, but does not account for **how much each feature actually varies in the data**. A feature with a large coefficient but near-zero variance will contribute very little to actual predictions in practice. The effect plot corrects for this by computing, for each instance and each feature, the effect:

$effect_{ij} = w_j \cdot x_{ij}$

The distribution of these effects across all instances is then displayed as a boxplot, ordered by mean effect to facilitate analysis.

Variance analysis of continuous features

Before plotting, we examine how much each continuous feature varies in the data:

Variable	Variance	Std. Dev.	Observation
days_since_2011	44591.00	211.17	Very high — spans 730 days
hum	202.86	14.24	Moderate variability
temp	56.33	7.51	Moderate-low variability
windspeed	26.96	5.19	Low variability

Table 2. Variance and standard deviation of the continuous model features.

Interpretation of the effect plot

days_since_2011: This variable generates the highest positive mean effect in the dataset. Since it spans 730 days, later dates (2012) contribute thousands of additional rentals compared to the first days of 2011. This contribution reflects the robust organic growth of the service over time.

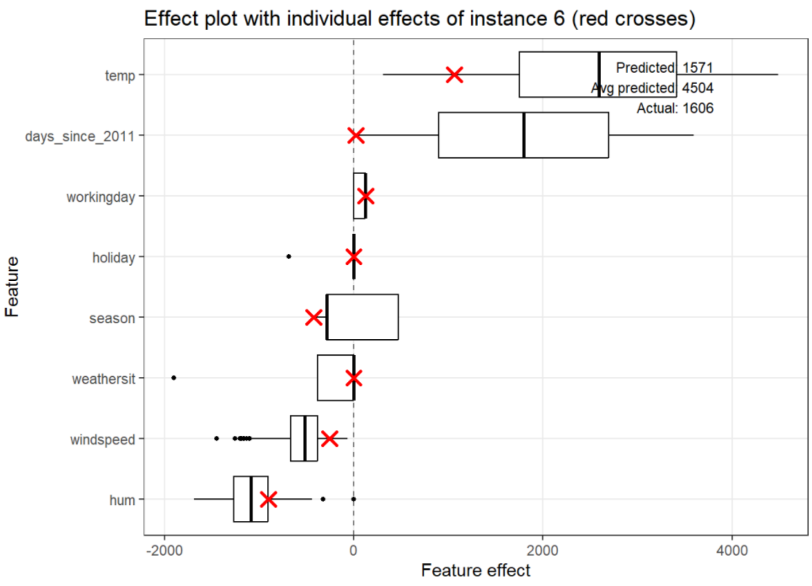
temp: Temperature shows the widest spread and the largest positive mean effect. Days at the cold end of the scale contribute almost zero (or slightly negative) effect, while warm summer days push this effect well above 2,000 additional rentals. This confirms temperature as one of the most operationally important drivers of demand.

windspeed: The effect of wind speed is negative but modest, and its distribution is relatively narrow. Even on the windiest days in the dataset, the dampening effect on rentals is far smaller than the boost from a warm day or the growth trend. This feature has lower business-critical importance than temperature or the temporal trend.

4.3. Explaining Individual Contributions

The effect plot can also be used to **explain a single prediction**, a technique known as local interpretability. By overlaying the feature effects of a specific instance on the global distribution, we can see whether that day's characteristics placed it above or below the typical effect range for each feature.

We analyse the prediction for the 6th instance in the dataset.



Key data for the 6th instance

- **Date:** 6 January 2011 (one of the earliest days in the dataset).
- **Predicted value:** 1,571 rentals.
- **Average predicted value (all days):** 4,504 rentals.
- **Actual value:** 1,606 rentals.

The model predicted **1,571 rentals** against the actual value of **1,606**, a reasonable approximation with an error of only 35 units. This instance sits well below the system-wide average (4,504 rentals), which can be fully explained by decomposing the contribution of each feature:

Feature contribution breakdown

- **days_since_2011:** Being one of the very first days in the dataset (day 5), the trend term contributes very little positive effect — far below the global median. This single factor already explains much of why the prediction falls so far below average.
- **temp:** January is cold in Washington D.C., so the temperature effect for this instance sits in the lower tail of the global distribution, contributing a small or mildly negative boost relative to the annual average.
- **hum and windspeed:** These features contribute modestly and negatively to the prediction, consistent with typical winter weather patterns.
- **season, weathersit, workingday, holiday:** The red crosses for these binary or categorical features fall very close to zero in the effect plot, indicating that their individual contribution to this particular instance is marginal.

Taken together, the model's prediction for day 6 is coherent and fully explainable: an early winter day in the first weeks of the service, with cold temperatures and normal conditions, generates far fewer rentals than the annual average. The effect decomposition allows us to **communicate this reasoning clearly and quantitatively** to any stakeholder, technical or otherwise.

5. Conclusions

The analysis of the Capital Bikeshare dataset through a multiple linear regression model yields several actionable insights for the company:

- **Weather and temperature are the most operationally relevant factors.** Temperature has one of the strongest positive effects on daily rentals, while rain and storms present the most significant deterrent (-1,901 rentals on a heavy-rain day). Although the company cannot control the weather, this finding justifies investing in demand forecasting systems that use weather predictions to optimise fleet repositioning, maintenance scheduling, and staffing levels.

- **The service is on a strong growth trajectory.** The days_since_2011 variable captures a robust upward trend: even after controlling for weather and seasonality, each additional day brings approximately 5 more rentals. This organic growth supports decisions on fleet expansion and infrastructure investment.

- **Commuter behaviour is a core pillar of demand.** Working days generate significantly more rentals than non-working days, while public holidays see a notable drop (-686 rentals). This profile is consistent with a user base dominated by regular commuters rather than leisure cyclists. Targeted promotions on holidays or weekends could help smooth demand and attract a more recreational segment.

- **Individual predictions are fully explainable, feature by feature.** The effect plot demonstrates that the model's predictions are completely decomposable: for any given day, it is possible to quantify exactly how much each variable contributed to the final forecast. This level of transparency is essential when communicating model outputs to non-technical stakeholders and justifying decisions based on algorithmic predictions.

- **The linear model has limitations that must be acknowledged.** The absence of interaction terms prevents the model from capturing conditional relationships (such as the differential effect of wind depending on day type). For greater predictive accuracy, more flexible models should be explored; however, linear regression remains indispensable as an interpretable and fully auditable baseline.

This analysis demonstrates that interpretability and practical utility are not mutually exclusive. A well-constructed and correctly interpreted linear model can deliver as much business value as more complex black-box models, with the additional advantage of being completely auditable and understandable by any stakeholder.